

Mr. Qihua Dong (His/He/Him)

Github: <https://github.com/dddraxxx>

Research Interests: **Computer Vision**

Seeking for **Continuous PhD's Supervision**

Education

City University of Hong Kong, Hong Kong SAR, China

Sep 2017 - Jul 2021

- BSc in Computer Science with Mathematics Minor; overall GPA 3.89/4.3 (Top 10% official proofed)
- QS Ranking 53rd, World University Rankings (General); ranking 78th in Computer Science and Information Systems (Ranking by subject)
- Featured courses: Enhanced Calculus and Linear Algebra; Data Structures; Discrete Mathematics; Statistic Processes; Advanced Programming; Design and Analysis of Algorithms; Real Analysis
- Dissertation topic: "The Impact of Task Scheduling on the Performance in Autonomous Driving Cars"

Publications

- He Ruozhen*, **Dong Qihua***, Lin Jiayin, Rynson Lau. "Weakly-Supervised Camouflaged Object Detection with Scribble Annotations". *Proc. AAAI*, 2023.
Paper link: <https://arxiv.org/pdf/2207.14083v2.pdf>
- Du Hao*, **Dong Qihua***, Xu Yan, Liao Jing. "Weakly-Supervised 3D Medical Image Segmentation using Geometric Prior and Contrastive Similarity". Under Major Review of *IEEE Transactions on Medical Imaging*, 2023.
- Du Hao*, **Dong Qihua***, Xu Yan, Liao Jing. "TDFormer: Top-Down Token Generation for 3D Medical Segmentation". Submitted to *Proc. IEEE CVPR*, 2023.
- **DONG Qihua**, Liang Yongqing, Li Xin, Wang Wenping. "Editable Skull to 3D Face Reconstruction via 2D Generative Model". Prepared to be submitted to *Proc. Siggraph*, 2023.
- **Dong Qihua**, Xu Yan, Liao Jing. "Volume-Preserving Registration for 3D Medical Images with Tumors". Prepared to be submitted to *Proc. ICCV*, 2023.

*: The authors with * contributed equally to the work

Honours and Scholarship

2021	First class honor , Bachelor's degree of City University of Hong Kong, HK
2019	Bronze Medal , The 2019 ICPC Asia Nanchang Regional Contest, Jiangxi Normal University, China
2017 - 2021	Dean's List , City University of Hong Kong, HK Sem A and B Year 1, Sem B Year 2, Sem B Year 3, Sem A Year 4
2017 - 2021	Full Tuition Scholarship , City University of Hong Kong, HK

Research Experience

Skull to 3D Face Reconstruction via 2D Generative Adversarial Model

Oct 2022 - Present

- Supervisor: Prof. Wenping Wang
- Research brief: Designed a 2D generative model to provide new potential portrait of the given skull and reconstruct the 3D face model from the 2D face image; the reconstruction can be manipulated to render faces from different groups of people, increasing the flexibility of the model in forensic applications.

Preserving Tumor Volume in 3D Medical Image Registration

July 2022 - Present

- Supervisor: Prof Jing Liao, Prof Yan Xu
- Research brief: Proposed a novel registration neural network that preserves the volume of tumors without the loss of alignment accuracy, allowing registration network to have wider clinical application. A new volume-preserving loss is designed specifically for the task.

Dynamic Transformer for 3D Medical Image Segmentation

June 2022 – Nov 2022

- Supervisor: Prof Jing Liao, Prof Yan Xu
- Research brief: Designed an efficient and accurate transformer that adapts to medical image segmentation with dynamic token generation, in which only foreground-related tokens are further split and attended. Our method improves SOTA Dice scores of transformers on two popular datasets in medical images.

Weakly-Supervised 3D Medical Image Segmentation Framework

Feb 2022 - May 2022

- Supervisor: Prof Jing Liao, Prof Yan Xu
- Research brief: Designed a new weakly supervised framework to segment medical images with bounding-box labels. The framework exploited the geometric prior of organs by using template organs in point-cloud form, which provides delicate shape guidance for segments. Besides, pretraining of the feature head was performed to learn contrasted features between objects and background. The work improves SOTA performance by 12.5% on Dice on two popular datasets in medical images.

Weakly-Supervised Camouflaged Object Detection

Dec 2021 - June 2022

- Supervisor: Prof Rynson Lau
- Research brief: Proposed the first weakly-supervised task and dataset in camouflaged object detection. It contains the novel consistency loss that corrects an ignored bias in previous works. The feature loss leveraging semantic features to infer concealed objects. Our model outperforms SOTA methods on COD benchmarks with an average improvement of 11.0% on MAE.

Swin Transformer on 3D Medical Image

Aug 2021 - Jan 2022

- Supervisor: Prof Jing Liao, Prof Yan Xu
- Research brief: Took advantage of the Swin architecture to conduct medical image segmentation.

Work Experience

Texas A&M University, TX, USA

Oct 2022 - Present

- Supervisor: Prof Wenping Wang (<https://engineering.tamu.edu/cse/profiles/Wang-Wenping.html>)
- Position: Research Assistant (Remote)
- Responsibilities: Conducting research related to faces in computer vision and computer graphics.

City University of Hong Kong, Hong Kong SAR, China

Aug 2021 - Present

- Supervisor: Prof Jing Liao (<https://www.cityu.edu.hk/stfprofile/jingliao.htm>)
- Position: Research Assistant (Full-time)
- Responsibilities: Engaging the research projects in computer vision, the primary research domain is 3D Medical Image Analysis and Weakly-Supervised Learning.

City University of Hong Kong, Hong Kong SAR, China

Sep 2020- Feb 2021

- Supervisor: Prof Jianping Wang (<https://www.cs.cityu.edu.hk/~jianwang/>)
- Position: Student Research Assistant (Part-time)
- Responsibilities: Research works in computer science: the major research topics are related to system scheduling schemes for self-driving tasks.

Sik Sik Yuen, Hong Kong SAR, China

Sep 2019- Jun 2020

- Position: Full-stack Programmer (Intern)
- Responsibilities: Mobile phone application development project, adding new functions which identify the emotion of users as well as maintaining the existing application and web service.

Skills

- Computer languages Python, C++, Matlab, R, Javascript, Java, CSS
- Tools and library PyTorch, Numpy, Detectron2, Skimage, mmdet, Monai, Scikit-learn,
- English skills TOEFL 110 (Nov 2022); GRE 330 (+3.5)